

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 128

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

When the Serial0/0/0 port is set as passive on R2, R2 cannot establish adjacency with R3; routing to R2 from R3 require R1 as the route, which increases the cost metric.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

no passive-interface S0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

__cost: 64, because only one link is used.

O (R2 Gigabit 0/0)

Is R2 listed as an OSPF neighbor to R3? _Yes_____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

__Because if the calculated cost is less than 1, the result will be rounded to 1, then the calculated cost is not accurate enough. Changing the reference bandwidth will fix this.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__Summing up all costs from every link in the route, the cost of each link is calculated by the formula: $\text{cost} = \text{ref bandwidth} / \text{interface bandwidth}$

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

__If the interface bandwidth is changed to 128 for all serial interfaces, 128 should be plugged in for "interface bandwidth" when calculating the costs for all interfaces instead of the default bandwidth.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

_Because the cost of the interface has increased. OSPF prefers routes with lowest costs.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

_Because every router in the OSPF protocol should have its unique ID, it is important for the Dijkstra algorithm.

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2. Why is the DR/BDR election process not a concern in this lab?

_Because the network topology is simple in this lab, there is no need to elect DR or BDR for broadcasting.

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3. Why would you want to set an OSPF interface to passive?

_Sometimes the network administrator might want to disable certain routing schemes for various purposes.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

_OSPF divides routers into areas that connect to a backbone, this allows easier management of big networks.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

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_network 10.0.0.0 0.255.255.255 area 0

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6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

_OSPF will be used. Because OSPF has the least administrative distance.

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